FIBRES AND LOCAL-RING FIBRES AS TOPOLOGICAL FIBRE SQUARES

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Editorial note. Counted span is the author's own proof environment, source lines 3751-3759, following the statement at 3727-3749.

Editorial note. The general case 'follows by covering X by open affines' patches the affine statement over the cover; the two diagrams in the statement are handled by the same argument.

Allowed standard prerequisites (editorial).

- Morphisms from an affine scheme of a local ring. lemma-morphism-from-spec-local-ring (card L26396, cited at ~3754). Bottom arrows factor through any affine U containing s.
- Fundamental diagram for maps of affines. Algebra, Remark algebra-remark-fundamental-diagram (cross-chapter, tag index verified, cited at ~3757). Affine case: fibre squares of spectra are topological fibre squares.

Source reference. Original reference 00E6: focused-seeds/geometry/S9-schemes-fibre-topological/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01J6: sources/stacks-schemes/pinned-original-source.tex, source lines 2550–2550. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Consider the diagrams*

$$\begin{array}{ccc} X_s & \longrightarrow & X \\ \downarrow & & \downarrow \\ \mathrm{Spec}(\kappa(s)) & \longrightarrow & S \end{array} \qquad \text{and} \qquad \begin{array}{ccc} \mathrm{Spec}(\mathcal{O}_{S,s}) \times_S X & \longrightarrow & X \\ \downarrow & & \downarrow \\ \mathrm{Spec}(\mathcal{O}_{S,s}) & \longrightarrow & S \end{array}$$

In both cases these diagrams induce fibre squares of topological spaces and in particular, the top horizontal arrow is a homeomorphism onto its image.

Proof. Choose an open affine $U \subset S$ that contains s. The bottom horizontal morphisms factor through U, see Lemma 01J6 for example. Thus we may assume that S is affine. If X is also affine, then the result follows from Algebra, Remark 00E6. In the general case the result follows by covering X by open affines. □

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